

Cover Letter — Journal of Chemical Information and Modeling

August 10, 2026

Dear Editors,

We submit the manuscript entitled “*Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes: a computational analysis*” for consideration as a research article.

The manuscript presents an integrated computational study connecting African-natural-product-inspired chemical-space exploration with target breadth and mutation-aware resistance hypotheses. Its central methodological premise is that these properties must not be collapsed into a single score: target breadth is represented by a target-specific docking profile, whereas resistance resilience is computed separately from mutation panels using a per-target denominator that excludes non-binding wild-type baselines.

The workflow combines a hybrid library of 65 856 molecules with a computationally prioritized set of 19 913 synthesizability-filtered leads and a locked 17-member Set-C cohort. The cohort was evaluated against PfDHFR, PfCRT, PfClpP, and PfATP4 using target-specific structural anchors, yielding 68 raw Vina records that passed an automated geometric gate; the underlying poses are archived with full provenance in the repository for independent inspection. Because Vina scores are not calibrated across unlike receptor pockets, the manuscript reports target-wise evidence and does not use a cross-target arithmetic mean. Corrected RRS analysis was derived separately from the declared WT/mutant panel, rather than from the four-target wild-type matrix, and classified the cohort as six A*, five B, five C, and one D profiles. The cross-metric analyses were retained as exploratory and null findings were reported without reinterpretation.

The manuscript addresses a specific prioritization problem: how to combine target breadth and mutation resilience without treating either docking score as experimental activity. Its contribution is a target-specific computational analysis that separates these axes, identifies their joint and discordant profiles, and converts them into testable priorities for biochemical, cellular, mutant-panel, and molecular-dynamics follow-up. The study does not claim measured potency, confirmed target engagement, resistance circumvention, or experimental IC₅₀/EC₅₀.

Source code, candidate manifests, receptor and ligand files, raw Vina outputs, RRS tables, configuration files, and provenance records are available at https://github.com/NanaEngo/Malaria_codesV2. All authors have reviewed and approved the manuscript and declare no competing financial interest. This manuscript is not under consideration elsewhere.

Thank you for considering our work.

Sincerely,
Serge Guy Nana Engo
for all authors